

Exploratory Factor Analysis



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Estimating Latent/Hidden Factors From Data

ID	English Score	Maths Score	Physics Score	Chemistry Score	Humanities Score	Lab Exam Score	Sports Score	Movies Watched	Parents Separated
	ES	MS	PS	CS	HS	LS	SS	MW	PP
21	51	24	23	22	43	45	82	10	0
23	54	55	66	89	89	76	54	5	1
27	22	88	77	66	90	56	89	7	0
52	31	45	21	89	34	43	90	9	0
64	77	90	89	56	56	76	76	7	0
32	22	32	32	67	34	56	67	3	0
41	30	77	54	45	22	45	90	9	1

Hidden / Latent Factors

Studious (*st*), Intelligent (*ig*), Mental Stability (*ms*), Physical Fitness (*pf*), Disciplined (*ds*)

Estimating Latent/Hidden Factors From Data

$$\mathbf{x} = \begin{bmatrix} \text{English Score (ES)} \\ \text{Maths Score (MS)} \\ \text{Physics Score (PS)} \\ \text{Chemistry Score (CS)} \\ \text{Humanities Score (HS)} \\ \text{Lab Exam Score (LS)} \\ \text{Sports Score (SS)} \\ \text{Movies Watched (MW)} \\ \text{Parents Separated (PP)} \end{bmatrix}$$

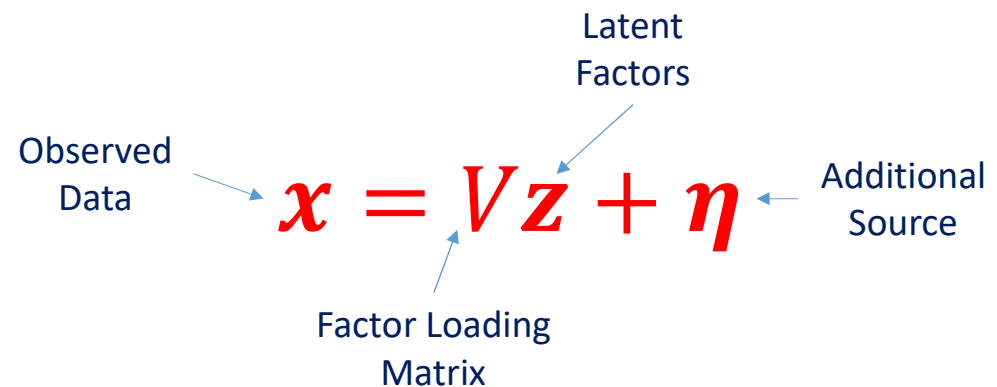
$$\mathbf{z} = \begin{bmatrix} \text{Studious (st)} \\ \text{Intelligent (ig)} \\ \text{Mental Stability (ms)} \\ \text{Physical Fitness (pf)} \\ \text{Disciplined (ds)} \end{bmatrix}$$

Observed Data $\rightarrow$ $\mathbf{x} = \mathbf{V}\mathbf{z} + \boldsymbol{\eta}$ $\leftarrow$ Additional Source

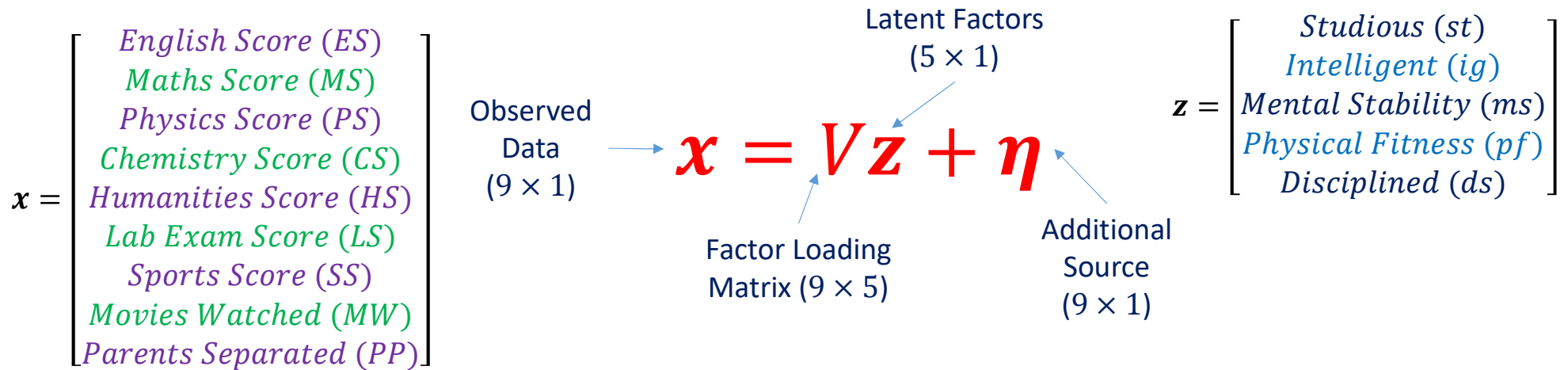
Latent Factors $\rightarrow$ $\mathbf{z}$

Factor Loading Matrix $\rightarrow$ $\mathbf{V}$

$\boldsymbol{\eta}$



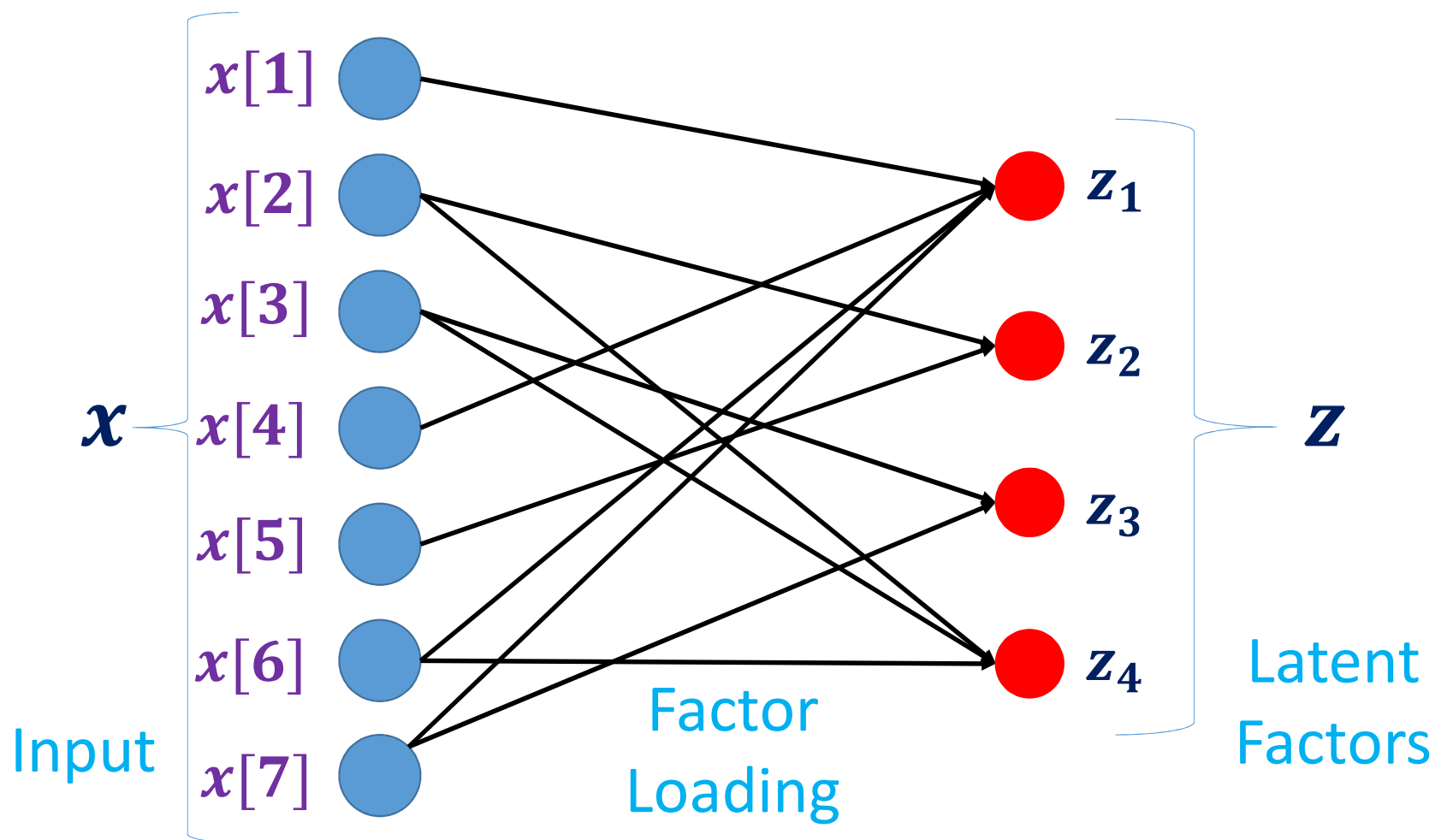
Estimating Latent/Hidden Factors From Data



$$\mathbf{x}[3] = \mathbf{V}[3][1] \times \mathbf{z}[1] + \mathbf{V}[3][2] \times \mathbf{z}[2] + \mathbf{V}[3][3] \times \mathbf{z}[3] + \mathbf{V}[3][4] \times \mathbf{z}[4] + \mathbf{V}[3][5] \times \mathbf{z}[5] + \boldsymbol{\eta}[3]$$

$$PS = \mathbf{V}[3][1] \times (st) + \mathbf{V}[3][2] \times (ig) + \mathbf{V}[3][3] \times (ms) + \mathbf{V}[3][4] \times (pf) + \mathbf{V}[3][5] \times (ds) + \boldsymbol{\eta}[3]$$

EFA: Input & Latent Factors



Factor Analysis

- Variables Depend on Latent Factors
- Factor Analysis Estimates Factor Loading
- Two Types of Factor Analysis
 - Exploratory Factor Analysis (EFA)
 - Confirmatory Factor Analysis (CFA)
- Provides Insight into Latent Factor Dependence
- Applications Involving Many Latent Factors
 - Started with Psychology
 - Useful in Economics & Sociology
 - Widely used in Speech Processing

Definitions & Assumptions: Input

Input: $\mathbf{x} \sim P_X(\mathbf{x}); \mathbf{x} \in \mathbb{R}^D$

$$E(\mathbf{x}) = \boldsymbol{\mu}$$

$$Cov(\mathbf{x}) = \mathbf{C}_{(D \times D)}$$

$$\tilde{\mathbf{x}} = \mathbf{x} - \boldsymbol{\mu}$$

$$E(\tilde{\mathbf{x}}) = \mathbf{0}$$

$$Cov(\tilde{\mathbf{x}}) = \mathbf{C}$$

Definitions & Assumptions: Latent Factor

Latent Factor: $\mathbf{z} \sim P_{\mathbf{z}}(\mathbf{z}); \mathbf{z} \in \mathbb{R}^M$

$$E(\mathbf{z}) = \mathbf{0}$$

$$Cov(\mathbf{z}) = \mathbf{I}_{(M \times M)}$$

$$E(\mathbf{z}[m]) = 0; m = 1, \dots, M$$

$$Var(\mathbf{z}[m]) = 1; m = 1, \dots, M$$

$$Cov(\mathbf{z}[j], \mathbf{z}[m]) = \delta[j - m] \quad \forall j, m$$

Kronecker Delta Function

$$\delta[i - j] = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

Definitions & Assumptions: Additional Source

Additional Source: $\boldsymbol{\eta} \sim P_{\mathbf{H}}(\boldsymbol{\eta}); \boldsymbol{\eta} \in \mathbb{R}^D$

$$E(\boldsymbol{\eta}) = \mathbf{0}$$

$$Cov(\boldsymbol{\eta}) = \mathbf{S}_{(D \times D)}$$

$$E(\boldsymbol{\eta}[i]) = 0; i = 1, \dots, D$$

$$Var(\boldsymbol{\eta}[i]) = \sigma_i^2; i = 1, \dots, D$$

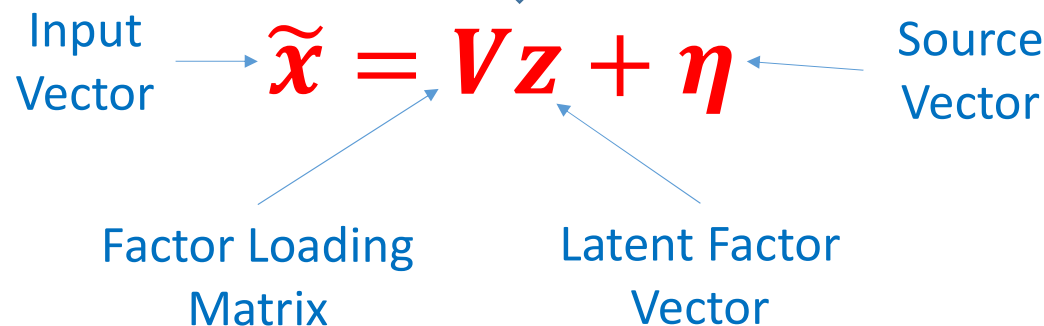
$$Cov(\boldsymbol{\eta}[i], \mathbf{z}[m]) = 0; \forall i, m$$

$$Cov(\boldsymbol{\eta}[i], \boldsymbol{\eta}[j]) = 0; i \neq j$$

EFA: Relating Input, Latent Factors and Source

$$\tilde{\mathbf{x}}[i] = \mathbf{V}[i][1]\mathbf{z}[1] + \cdots + \mathbf{V}[i][m]\mathbf{z}[m] + \cdots + \mathbf{V}[i][M]\mathbf{z}[M] + \boldsymbol{\eta}[i]$$


$$\tilde{\mathbf{x}}[i] = \sum_{m=1}^M \mathbf{V}[i][m]\mathbf{z}[m] + \boldsymbol{\eta}[i]$$



Input Vector $\rightarrow \tilde{\mathbf{x}} = \mathbf{V}\mathbf{z} + \boldsymbol{\eta}$ Source Vector

Factor Loading Matrix $\rightarrow \mathbf{V}$ Latent Factor Vector $\rightarrow \mathbf{z}$

Recapitulation: On Variance

$$\gamma = a\theta + b\phi$$

$$\text{Cov}(\theta, \phi) = E[\{\theta - E(\theta)\}\{\phi - E(\phi)\}] = 0$$

$$\text{Var}(\gamma) = E[\{\gamma - E(\gamma)\}^2] = ?$$

Recapitulation: Mathematical Expectation

Consider Random Variable x with Density/Mass Function $p(x)$

$$E[g(X)] = \int_{-\infty}^{\infty} g(X)p(X)dX \qquad E[h(X)] = \sum_i h(X_i)p(X_i)$$

Linear Operator: $E[a_1g(X) + a_2h(X)] = a_1E[g(X)] + a_2E[h(X)]$

Operating on a Constant K : $E[K] = K$

$E[X]$: *Mean*

$E[(X - \mu)^2]$: *Variance*

Recapitulation: On Variance

$$\gamma = a\theta + b\phi; \text{Cov}(\theta, \phi) = 0$$

$$\text{Var}(\gamma) = E[\{(a\theta + b\phi) - E(a\theta + b\phi)\}^2]$$

$$= E[\{(a\theta + b\phi) - E(a\theta + b\phi)\}^2]$$

$$= E[\{(a\theta - aE(\theta)) + (b\phi - bE(\phi))\}^2]$$

$$= E[\{a\theta - aE(\theta)\}^2 + 2\{a\theta - aE(\theta)\}\{b\phi - bE(\phi)\} + \{b\phi - bE(\phi)\}^2]$$

$$= a^2 E\{\theta - E(\theta)\}^2 + 2ab E\{\theta - E(\theta)\}\{\phi - E(\phi)\} + b^2 E\{\phi - E(\phi)\}^2$$

$$\text{Var}(\gamma) = a^2 \text{Var}(\theta) + b^2 \text{Var}(\phi)$$

Relating Input, Factor Loading and Source Variance

$$\tilde{\mathbf{x}}[i] = \sum_{m=1}^M \mathbf{v}[i][m] \mathbf{z}[m] + \boldsymbol{\eta}[i]$$

$$\text{Var}(\tilde{\mathbf{x}}[i]) = ?$$

$$\text{Var}(\mathbf{z}[m]) = 1; m = 1, \dots, M$$

Relating Input, Factor Loading and Source Variance

$$\tilde{\mathbf{x}}[i] = \sum_{m=1}^M \mathbf{v}[i][m] \mathbf{z}[m] + \boldsymbol{\eta}[i]$$



$$\text{Var}(\tilde{\mathbf{x}}[i]) = \sum_{m=1}^M \text{Var}(\mathbf{v}[i][m] \mathbf{z}[m]) + \text{Var}(\boldsymbol{\eta}[i])$$



$$\text{Var}(\tilde{\mathbf{x}}[i]) = \sum_{m=1}^M (\mathbf{v}[i][m])^2 + \sigma_i^2$$

$$\text{Var}(\mathbf{z}[m]) = 1; m = 1, \dots, M$$

Relating Covariance of Input, Factor and Source

$$\tilde{\mathbf{x}} = \mathbf{V}\mathbf{z} + \boldsymbol{\eta}$$



$$\text{Cov}(\tilde{\mathbf{x}}) = ?$$

$$\text{Cov}(\mathbf{z}) = \mathbf{I}$$

$$\text{Cov}(\boldsymbol{\eta}) = \mathbf{S}$$

Relating Covariance of Input, Factor and Source

$$\begin{aligned} \text{Cov}(\tilde{\mathbf{x}}) &= E[\{\tilde{\mathbf{x}} - E(\tilde{\mathbf{x}})\}\{\tilde{\mathbf{x}} - E(\tilde{\mathbf{x}})\}^T] = E[(\tilde{\mathbf{x}})(\tilde{\mathbf{x}})^T] \\ &= E[(V\mathbf{z} + \boldsymbol{\eta})(V\mathbf{z} + \boldsymbol{\eta})^T] \\ &= E[V\mathbf{z}\mathbf{z}^T V^T + V\mathbf{z}\boldsymbol{\eta}^T + \boldsymbol{\eta}\mathbf{z}^T V^T + \boldsymbol{\eta}\boldsymbol{\eta}^T] \\ &= E(V\mathbf{z}\mathbf{z}^T V^T) + E(V\mathbf{z}\boldsymbol{\eta}^T) + E(\boldsymbol{\eta}\mathbf{z}^T V^T) + E(\boldsymbol{\eta}\boldsymbol{\eta}^T) \\ &= VE(\mathbf{z}\mathbf{z}^T)V^T + VE(\mathbf{z}\boldsymbol{\eta}^T) + E(\boldsymbol{\eta}\mathbf{z}^T)V^T + E(\boldsymbol{\eta}\boldsymbol{\eta}^T) \\ &= VCov(\mathbf{z})V^T + V * \mathbf{0}_{(m \times d)} + \mathbf{0}_{(d \times m)} * V^T + Cov(\boldsymbol{\eta}) \end{aligned}$$

$$\text{Cov}(\tilde{\mathbf{x}}) = VV^T + S$$

$$E(\tilde{\mathbf{x}}) = \mathbf{0}$$

$$E(\mathbf{z}) = \mathbf{0}$$

$$E(\boldsymbol{\eta}) = \mathbf{0}$$

$$E(\boldsymbol{\eta}[i]\mathbf{z}[m]) = 0$$

$$Cov(\mathbf{z}) = I$$

$$Cov(\boldsymbol{\eta}) = S$$

Recapitulation: On Eigen Values & Vectors

$$A = (\underbrace{e_1 \dots e_m}_{\text{Matrix Block for Top-m Eigen Values and Corresponding Eigen Vectors}} \dots e_n) \begin{pmatrix} \lambda_1 & \dots & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_m & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \dots & \lambda_n \end{pmatrix} \begin{pmatrix} e_1^T \\ \vdots \\ e_m^T \\ \vdots \\ e_n^T \end{pmatrix}$$

Matrix Block for Top-m Eigen Values and Corresponding Eigen Vectors

$$A = (\underbrace{E_1 \quad E_2}_{\text{Plays Main Role in Reconstructing } A}) \begin{pmatrix} \Lambda_1 & \mathbf{0} \\ \mathbf{0} & \Lambda_2 \end{pmatrix} \begin{pmatrix} E_1^T \\ E_2^T \end{pmatrix} = \underbrace{E_1 \Lambda_1 E_1^T}_{\text{Has Comparatively Small Value and can be Ignored}} + \underbrace{E_2 \Lambda_2 E_2^T}_{\text{Has Comparatively Small Value and can be Ignored}}$$

Estimating Factor Loading Matrix

Decomposition for
Lower Eigen Values

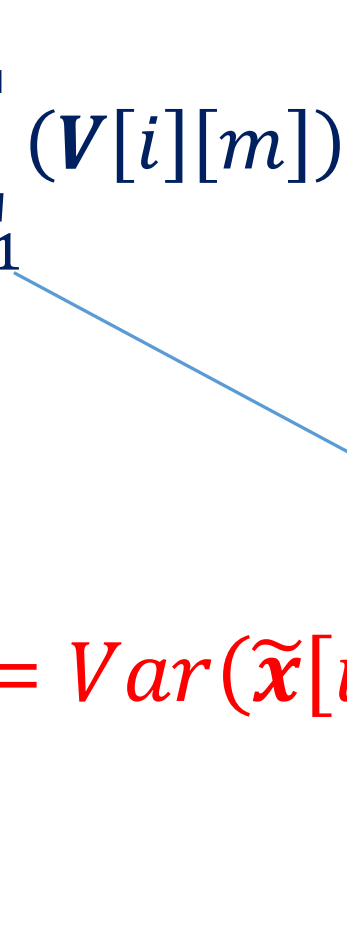
$$\text{Cov}(\tilde{\mathbf{x}}) = \mathbf{C} = \mathbf{E}\mathbf{Q}\mathbf{E}^T = \mathbf{E}_1\mathbf{\Lambda}_1\mathbf{E}_1^T + \mathbf{E}_2\mathbf{\Lambda}_2\mathbf{E}_2^T$$

Decomposition for
Top-M Eigen Values

$$\mathbf{V}\mathbf{V}^T \approx \mathbf{E}_1\mathbf{\Lambda}_1\mathbf{E}_1^T = \mathbf{E}_1\sqrt{\mathbf{\Lambda}_1}\sqrt{\mathbf{\Lambda}_1}\mathbf{E}_1^T = \{\mathbf{E}_1\sqrt{\mathbf{\Lambda}_1}\}\{\mathbf{E}_1\sqrt{\mathbf{\Lambda}_1}\}^T$$

$$\mathbf{V} = \mathbf{E}_1\sqrt{\mathbf{\Lambda}_1}$$

Estimating the Source

$$Var(\tilde{\mathbf{x}}[i]) = \sum_{m=1}^M (\mathbf{V}[i][m])^2 + \sigma_i^2 \quad \mathbf{V} = \mathbf{E}_1 \sqrt{\Lambda_1}$$


$$\sigma_i^2 = Var(\tilde{\mathbf{x}}[i]) - \sum_{m=1}^M (\mathbf{V}[i][m])^2$$

EFA: The Steps

$$X = \{\mathbf{x}_n : \mathbf{x}_n \in \mathbb{R}^D; n = 1, \dots, N\}$$

$$\hat{\boldsymbol{\mu}} = \text{Mean}(X)$$

$$\hat{\mathbf{C}} = \text{Cov}(X)$$

$$\tilde{X} = \{\tilde{\mathbf{x}}_n : \tilde{\mathbf{x}}_n = \mathbf{x}_n - \hat{\boldsymbol{\mu}}; \mathbf{x}_n \in X\}$$

Factor
Analysis

$$\{\hat{\mathbf{V}}, \hat{\mathbf{S}}\}$$

EFA: Numerical Example

#	1	2	3	4	5	6	7	8	9	10
R_1	11	3	6	16	0	16	0	14	18	13
R_2	8	13	16	8	1	8	7	19	14	19
R_3	4	3	13	9	4	1	15	14	13	18

Input Dataset: Three Readings of Ten Subjects

EFA: Evaluating Mean and Covariance Matrix

#	1	2	3	4	5	6	7	8	9	10
R ₁	11	3	6	16	0	16	0	14	18	13
R ₂	8	13	16	8	1	8	7	19	14	19
R ₃	4	3	13	9	4	1	15	14	13	18

$$\boldsymbol{\mu} = \begin{bmatrix} 9.7 \\ 11.3 \\ 9.4 \end{bmatrix}$$

$$\boldsymbol{C} = \begin{bmatrix} 42.61 & 14.79 & 4.32 \\ 14.79 & 30.81 & 19.58 \\ 4.32 & 19.58 & 32.24 \end{bmatrix}$$

EFA: Zero-Centering Input Data

#	1	2	3	4	5	6	7	8	9	10
R ₁	1.3	-6.7	-3.7	6.3	-9.7	6.3	-9.7	4.3	8.3	3.3
R ₂	-3.3	1.7	4.7	-3.3	-10.3	-3.3	-4.3	7.7	2.7	7.7
R ₃	-5.4	-6.4	3.6	-0.4	-5.4	-8.4	5.6	4.6	3.6	8.6

$$\boldsymbol{\mu} = \begin{bmatrix} 9.7 \\ 11.3 \\ 9.4 \end{bmatrix}$$

$$\boldsymbol{C} = \begin{bmatrix} 42.61 & 14.79 & 4.32 \\ 14.79 & 30.81 & 19.58 \\ 4.32 & 19.58 & 32.24 \end{bmatrix}$$

EFA: Decomposing Covariance Matrix

$$\mathbf{C} = \begin{bmatrix} 42.61 & 14.79 & 4.32 \\ 14.79 & 30.81 & 19.58 \\ 4.32 & 19.58 & 32.24 \end{bmatrix}$$



$$\mathbf{C} = \begin{bmatrix} -0.603 & 0.754 & -0.259 \\ -0.617 & -0.236 & 0.751 \\ -0.505 & -0.613 & -0.607 \end{bmatrix} \begin{bmatrix} 61.32 & 0 & 0 \\ 0 & 34.46 & 0 \\ 0 & 0 & 9.87 \end{bmatrix} \begin{bmatrix} -0.603 & 0.754 & -0.259 \\ -0.617 & -0.236 & 0.751 \\ -0.505 & -0.613 & -0.607 \end{bmatrix}^T$$

Three Readings of Ten Subjects AND Two Latent Factors

EFA: Decomposing Covariance Matrix

$$\mathbf{C} = \begin{bmatrix} 42.61 & 14.79 & 4.32 \\ 14.79 & 30.81 & 19.58 \\ 4.32 & 19.58 & 32.24 \end{bmatrix} = \mathbf{E}_1 \mathbf{\Lambda}_1 \mathbf{E}_1^T + \mathbf{E}_2 \mathbf{\Lambda}_2 \mathbf{E}_2^T$$

$$\mathbf{E}_1 = \begin{bmatrix} -0.603 & 0.754 \\ -0.617 & -0.236 \\ -0.505 & -0.613 \end{bmatrix}$$

$$\mathbf{\Lambda}_1 = \begin{bmatrix} 61.32 & 0 \\ 0 & 34.46 \end{bmatrix}$$

$$\mathbf{E}_2 = \begin{bmatrix} -0.259 \\ 0.751 \\ -0.607 \end{bmatrix}$$

$$\mathbf{\Lambda}_2 = [9.87]$$

EFA: Computing Factor Loading Matrix

$$V = E_1 \sqrt{\Lambda_1}$$

$$V = \begin{bmatrix} -0.603 & 0.754 \\ -0.617 & -0.236 \\ -0.505 & -0.613 \end{bmatrix} \sqrt{\begin{bmatrix} 61.32 & 0 \\ 0 & 34.46 \end{bmatrix}} = \begin{bmatrix} -0.603 & 0.754 \\ -0.617 & -0.236 \\ -0.505 & -0.613 \end{bmatrix} \begin{bmatrix} \sqrt{61.32} & 0 \\ 0 & \sqrt{34.46} \end{bmatrix}$$

$$V = \begin{bmatrix} -4.7295 & -4.4248 \\ -4.8294 & 1.3852 \\ -3.9537 & 3.6008 \end{bmatrix}$$

EFA: Computing Additional Source Variances

$$\mathbf{C} = \begin{bmatrix} 42.61 & 14.79 & 4.32 \\ 14.79 & 30.81 & 19.58 \\ 4.32 & 19.58 & 32.24 \end{bmatrix}$$

$$\sigma_i^2 = \text{Var}(\tilde{\mathbf{x}}[i]) - \sum_{m=1}^M (\mathbf{V}[i][m])^2$$

$$\mathbf{V} = \begin{bmatrix} -4.7295 & -4.4248 \\ -4.8294 & 1.3852 \\ -3.9537 & 3.6008 \end{bmatrix}$$

$$\sigma_1^2 = 0.6628$$

$$\sigma_2^2 = 5.5676$$

$$\sigma_3^2 = 3.6413$$

EFA: Dimensionality Reduction Transformation

$$\mathbf{z}_n = \mathbf{W}\tilde{\mathbf{x}}_n$$

Dimensionality Reduction
Transformation Relation

Construct Following Matrices by Taking All Samples

$$\mathbf{Y} = [\tilde{\mathbf{x}}_1 \quad \cdots \quad \tilde{\mathbf{x}}_n \quad \cdots \quad \tilde{\mathbf{x}}_N]$$

$$\mathbf{Z} = [\mathbf{z}_1 \quad \cdots \quad \mathbf{z}_n \quad \cdots \quad \mathbf{z}_N]$$

$$\mathbf{Z} = \mathbf{W}\mathbf{Y}$$

EFA: Solving for W

$$\mathbf{Z} = \mathbf{W}\mathbf{Y}$$



$$\mathbf{W} = (\mathbf{Z}\mathbf{Y}^T)(\mathbf{Y}\mathbf{Y}^T)^{-1}$$

Generalized Moore-Penrose Pseudo-inverse

Evaluating YY^T

$$Y = [\tilde{\mathbf{x}}_1 \quad \cdots \quad \tilde{\mathbf{x}}_N]$$

$$Y^T = \begin{bmatrix} \tilde{\mathbf{x}}_1^T \\ \vdots \\ \tilde{\mathbf{x}}_N^T \end{bmatrix}$$

$$YY^T = \sum_{n=1}^N \tilde{\mathbf{x}}_n \tilde{\mathbf{x}}_n^T = N\hat{\mathbf{C}}$$

Covariance of Input and Factor: $Cov(\tilde{\mathbf{x}}, \mathbf{z})$

$$\begin{aligned} Cov(\tilde{\mathbf{x}}[i], \mathbf{z}[j]) &= \sum_{m=1}^M Cov(\mathbf{V}[i][m]\mathbf{z}[m], \mathbf{z}[j]) + Cov(\boldsymbol{\eta}[i], \mathbf{z}[j]) \\ &= \sum_{m=1}^M \mathbf{V}[i][m]Cov(\mathbf{z}[m], \mathbf{z}[j]) + Cov(\boldsymbol{\eta}[i], \mathbf{z}[j]) = \sum_{m=1}^M \mathbf{V}[i][m]\delta[j - m] \end{aligned}$$

$$Cov(\tilde{\mathbf{x}}[i], \mathbf{z}[j]) = \mathbf{V}[i][j]$$

$$Cov(\tilde{\mathbf{x}}, \mathbf{z}) = \mathbf{V}$$

Evaluating $\mathbf{ZY}^T$

$$\mathbf{Z} = [\mathbf{z}_1 \quad \cdots \quad \mathbf{z}_N]$$

$$\mathbf{Y}^T = \begin{bmatrix} \tilde{\mathbf{x}}_1^T \\ \vdots \\ \tilde{\mathbf{x}}_N^T \end{bmatrix}$$

$$\mathbf{ZY}^T = \sum_{n=1}^N \mathbf{z}_n \tilde{\mathbf{x}}_n^T = N \hat{\mathbf{V}}^T$$

EFA: Solving for W

$$W = (ZY^T)(YY^T)^{-1}$$



$$W = (N\hat{V}^T)(N\hat{C})^{-1}$$



$$W = \hat{V}^T \hat{C}^{-1}$$

EFA: Dimensionality Reduction

#	1	2	3	4	5	6	7	8	9	10
R ₁	-1.3	-6.7	-3.7	6.3	-9.7	6.3	-9.7	4.3	8.3	3.3
R ₂	-3.3	1.7	4.7	-3.3	-10.3	-3.3	-4.3	7.7	2.7	7.7
R ₃	-5.4	-6.4	3.6	-0.4	-5.4	-8.4	5.6	4.6	3.6	8.6

$$V = \begin{bmatrix} -4.7295 & -4.4248 \\ -4.8294 & 1.3852 \\ -3.9537 & 3.6008 \end{bmatrix}$$

$$C = \begin{bmatrix} 42.61 & 14.79 & 4.32 \\ 14.79 & 30.81 & 19.58 \\ 4.32 & 19.58 & 32.24 \end{bmatrix}$$

EFA: Computing Dimensionality Reduction Transform

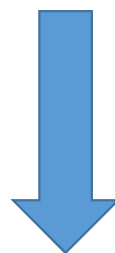
$$W = \hat{V}^T \hat{C}^{-1}$$

$$W = \begin{bmatrix} -4.7295 & -4.4248 \\ -4.8294 & 1.3852 \\ -3.9537 & 3.6008 \end{bmatrix}^T \begin{bmatrix} 42.61 & 14.79 & 4.32 \\ 14.79 & 30.81 & 19.58 \\ 4.32 & 19.58 & 32.24 \end{bmatrix}^{-1}$$

$$W = \begin{bmatrix} -0.077 & -0.079 & -0.064 \\ -0.128 & 0.040 & 0.104 \end{bmatrix}$$

EFA: Computing Factors

#	1	2	3	4	5	6	7	8	9	10
R ₁	-1.3	-6.7	-3.7	6.3	-9.7	6.3	-9.7	4.3	8.3	3.3
R ₂	-3.3	1.7	4.7	-3.3	-10.3	-3.3	-4.3	7.7	2.7	7.7
R ₃	-5.4	-6.4	3.6	-0.4	-5.4	-8.4	5.6	4.6	3.6	8.6



$$\mathbf{z} = (\hat{\mathbf{V}}^T \hat{\mathbf{C}}^{-1}) \tilde{\mathbf{x}}$$

#	1	2	3	4	5	6	7	8	9	10
z ₁	0.51	0.8	-0.32	-0.2	1.91	0.32	0.73	-1.23	-1.08	-1.42
z ₂	-0.86	0.26	1.04	-0.98	0.27	-1.82	1.66	0.24	-0.58	0.78

Summary

- Exploratory & Confirmatory Factor Analysis
- Unsupervised Algorithm
- Interpretable Latent Factors
- Interpretability as Advantage over PCA
- Applications involving Meaningful Factors
 - Psychology & Social Sciences
 - Economic Modeling
 - Engineering Applications



Thank You